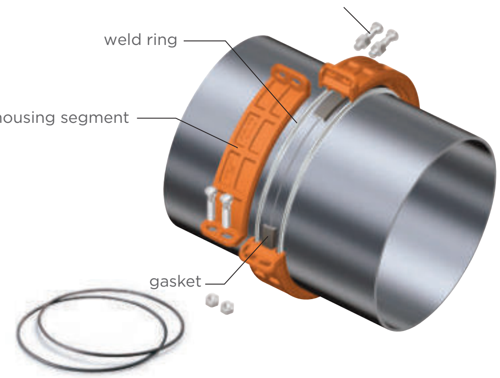
section 6

ring joint, shouldered & plain end couplings

Shurjoint ring joint, shouldered and plain end couplings are non-grooved mechanical pipe joining components and excellent alternatives where pipe is difficult to groove or when grooving is not the preferred method. The processing of a roll groove on pipe becomes more difficult as the pipe O.D. and or wall thickness increases. Roll grooving pipe larger than 14" (350 mm) requires proper tools and equipment. Pipe having a wall thickness greater than 0.375" (9.5 mm) may not be practical to roll groove.

Ring Joint Coupling

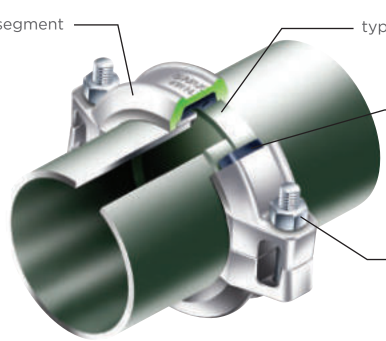
The Shurjoint ring joint coupling provides a much more secure joint than a comparable roll-grooved system, simply because the contact area of the rings is much greater than that of the roll-groove profile. In addition, the welded rings are able to withstand 2 - 3 times the shearing forces of roll grooves. High pressure couplings up to 3770 psi (260 Bar) are available.



#R-88 ring joint coupling

shouldered coupling

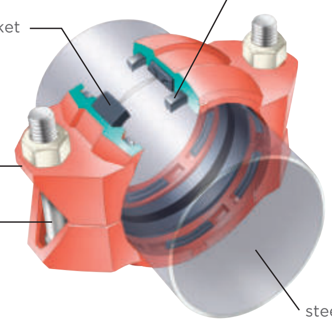
The Shurjoint shouldered piping system is a classic and versatile pipe joining method which utilizes Type A weld-on rings. Shurjoint shouldered couplings are widely used in irrigation, dewatering on construction sites, and other installations.



#S35 shouldered coupling

plain-end coupling

The Shurjoint plain-end coupling securely grips the pipe with its built-in case hardened jaws incorporated within heavy duty housing segments and heavy duty bolts and nuts. The Shurjoint Model 79 Wildcat coupling is designed to join plain-end or beveled-end carbon steel pipe without roll-grooving, welding or threading. The Model 79 plain-end coupling can be used for mining, process piping, manifold piping, and oil field services.



#79 plain-end coupling

R-88 ring joint coupling

typical applications

- water & waste water treatment plants
- mining & tunnel boring
- pulp & paper
- hydroelectric plants
- co-gen electric plants
- food & beverage
- compressed air
- HVAC

The Shurjoint Model R-88 Ring Joint Coupling is supplied with a pair of factory supplied weld rings. For installation weld a factory-supplied weld ring on each pipe end to be connected, next mount the rubber gasket over the pipe ends, place coupling segments over the gasket and fasten the bolts and nuts.

The Shurjoint R-88 Ring Joint Coupling is considered a shoulder coupling with the factory supplied weld rings serving as the joint shoulders. The R-88's performance standards meet and or exceed the requirements of ASTM F1476 and AWWA C606. The factory supplied weld rings offer a much more economical and convenient alternative to traditional shoulder rings, such as Type A, B, C, D, E, and G rings.

The R-88 Coupling can also be used on stainless steel pipe with optional weld rings available in compatible stainless steel grades. Check with Shurjoint for details and availability.

max. internal service pressures of carbon steel pipe

ASTM A53 Gr.B

When designing a piping system you must select pipe with the appropriate wall thickness to correspond with the intended working pressure of the system. The table lists design working pressure by the pipe wall schedule, XS, STD and LW, of representative ASTM A53 Gr. B carbon steel pipe calculated in accordance with the formula stipulated in ASME B31.1 Power Piping 104.1.

$$P = \frac{2SE\ (tm - A)}{Do - 2y\ (tm - A)}$$

where:

- **P** = Maximum internal service pressure (psi)
- **SE** = Allowable stress (psi) (ASTM A53 Gr. B = 15,000 psi)
- **tm** = Minimum pipe wall thickness (inch) (87.5% of nominal wall thickness)
- **Do** = Outside diameter of pipe (inch)
- **y** = A coefficient (For ferritic steels 600°F or below = 0.4)
- **A** = Additional thickness (inch) (A = 0)

max. internal service pressures of carbon steel pipe

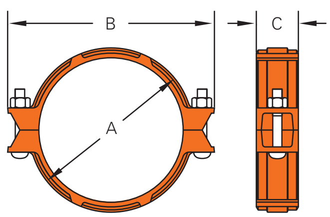
ASTM A53 Gr.B

nom. size (in/DN)	XS (0.5")	STD (0.375")	LW (0.25" / 0.312"^)
8 / 200	1586	1006	777
10 / 250	1262	913	621
12 / 300	1058	788	522
14 / 350	962	717	475
16 / 400	839	625	415
18 / 450	744	555	368
20 / 500	668	499	331
24 / 600	555	415	275
26 / 650	512	382	318
28 / 700	475	355	295
30 / 750	443	331	275
32 / 800	415	310	258
36 / 900	368	275	229
38 / 950	349	261	217
40 / 1000	331	248	206
42 / 1050	315	236	187
44 / 1100	301	225	-
48 / 1200	275	206	-
52 / 1300	254	190	-
54 / 1350	245	183	-
56 / 1400	236	177	-
60 / 1500	220	165	-
66 / 1650	200	150	-
68 / 1700	194	145	-
72 / 1800	183	137	-
84 / 2100	157	118	-
96 / 2400	137	103	-

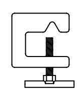
Except *8": 0.322", ^ 8" - 24": 0.25", 26" - 42": 0.312"

R-88 ring joint coupling
8" - 12"

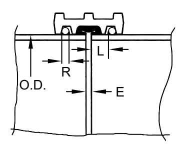
R-88
8" - 12"



8"~ 12"



Ring Clamp



The Shurjoint Model R-88 Ring Joint Coupling is an ideal pipe joining method when pipe is difficult to groove or when grooving is not the preferred joining method. Available in sizes 8" to 96" the R-88 offers ease of use and excellent performance.

dimensions

R88 (8" - 12")

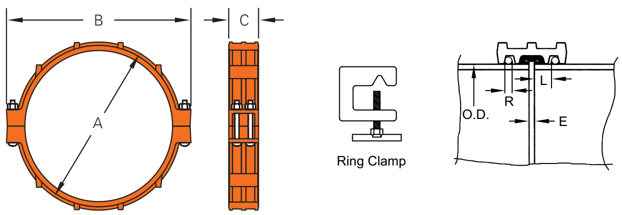
		rings both sides fully welded**			angular movement / deflection		dimensions			bolts					
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)*	axial displacement¹ E	per coupling	per pipe	A	B	C	no.	size	sealing surface L	ring size R	no. of clamps	weight
8" 200	8.625" (219.1 mm)	400 psi (28.0 bar)	23350 lbs (105.51 kN)	0-0.340" (0-8.7 mm)	2.14	0.45"/ft (37 mm/m)	10.08" (256 mm)	13.00" (330 mm)	3.11" (79 mm)	2	¾ x 4¾" (M20x120 mm)	0.91" (23 mm)	¼" (6.0 mm)	3	16.8 lbs (7.6 Kgs)
10" 250	10.750" (273.0 mm)	400 psi (28.0 bar)	36280 lbs (163.81 kN)	0-0.340" (0-8.7 mm)	1.95	0.41"/ft (34 mm/m)	12.29" (312 mm)	15.20" (386 mm)	3.25" (83 mm)	2	¾ x 4¾" (M20x120 mm)	0.91" (23 mm)	¼" (6.0 mm)	3	22.2 lbs (10.1 Kgs)
12" 300	12.750" (323.9 mm)	400 psi (28.0 bar)	51040 lbs (230.59 kN)	0-0.190" (0-4.8 mm)	0.82	0.17"/ft (14 mm/m)	14.72" (374 mm)	17.90" (455 mm)	3.39" (86 mm)	2	⅞ x 6½" ---	1.02" (26 mm)	5/16" (8.0 mm)	3	30.8 lbs (14.0 Kgs)
200 JIS	8.516" (216.3 mm)	400 psi (28.0 bar)	22770 lbs (102.83 kN)	0-0.340" (0-8.7 mm)	1.50	0.31"/ft (26 mm/m)	9.96" (253 mm)	12.87" (327 mm)	3.11" (79 mm)	2	--- (M20x120 mm)	0.91" (23 mm)	¼" (6.0 mm)	3	17.6 lbs (8.0 Kgs)
250 JIS	10.528" (267.4 mm)	400 psi (28.0 bar)	34800 lbs (157.16 kN)	0-0.340" (0-8.7 mm)	1.50	0.31"/ft (26 mm/m)	12.05" (306 mm)	14.96" (380 mm)	3.25" (83 mm)	2	--- (M20x120 mm)	0.91" (23 mm)	¼" (6.0 mm)	3	22.0 lbs (10.0 Kgs)
300 JIS	12.539" (318.5 mm)	400 psi (28.0 bar)	49360 lbs (222.97 kN)	0-0.190" (0-4.8 mm)	1.50	0.31"/ft (26 mm/m)	14.53" (369 mm)	17.72" (450 mm)	3.39" (86 mm)	2	--- (M20x120 mm)	1.02" (26 mm)	5/16" (8.0 mm)	3	32.6 lbs (14.8 Kgs)

1) Dimensions are subject to change without notice. Other sizes are available on request.
* Working Pressure is based on rings both sides fully welded standard wall carbon steel pipe.
\# Working Pressure and End Load are the total from all internal and external loads based on the applicable pipe wall thickness.
† Allowable Axial Displacement and Angular Movement (Deflection) figures shown are the maximum nominal range of movement at each R-88 coupling joint when rings are welded in the standard position. For design and installation purposes these figures should be reduced by 25%.
‡ The number of ring clamps listed is the minimum required to correctly position the weld ring around the circumference of the pipe end.

R-88N ring joint coupling

14" - 26"

R-88N
14" - 26"



14"~ 26"

The Shurjoint Model R-88N is a two segment Ring Joint Coupling. Available in sizes 14" to 26" (350 mm to 650 mm). The two-segment style offers an easier and faster installation.

dimensions

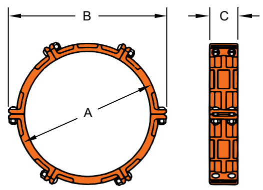
R88N (14" - 26")

		rings both sides fully welded**			angular movement / deflection		dimensions			bolts					
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)*	axial displacement[1] E	per coupling (°)	per pipe	A	B	C	no.	size	sealing surface L	ring size R	no. of clamps	weight
14" 350	14.000" (355.6 mm)	400 psi (28.0 bar)	61540 lbs (277.94 kN)	0-0.250" (0-6.4 mm)	1.20°	0.25"/ft (21 m/mm)	15.93" (405 mm)	19.40" (493 mm)	3.65" (93 mm)	2	⅞ x 5½" ---	1.02" (26 mm)	5/16" (8.0 mm)	4	31.5 lbs (14.3 Kgs)
16" 400	16.000" (406.4 mm)	400 psi (28.0 bar)	80380 lbs (363.02 kN)	0-0.250" (0-6.4 mm)	0.90°	0.19"/ft (16 m/mm)	17.92" (455 mm)	21.52" (547 mm)	3.65" (93 mm)	2	⅞ x 5½" ---	1.02" (26 mm)	5/16" (8.0 mm)	4	35.0 lbs (15.9 Kgs)
18" 450	18.000" (457.2 mm)	400 psi (28.0 bar)	101730 lbs (459.45 kN)	0-0.375" (0-9.5 mm)	1.20°	0.25"/ft (21 m/mm)	20.37" (517 mm)	24.17" (614 mm)	4.23" (107 mm)	2	1 x 5½" ---	1.18" (30 mm)	5/16" (8.0 mm)	4	59.9 lbs (27.2 Kgs)
20" 500	20.000" (508.0 mm)	400 psi (28.0 bar)	125600 lbs (567.22 kN)	0-0.375" (0-9.5 mm)	1.08°	0.23"/ft (19 m/mm)	22.46" (570 mm)	25.99" (660 mm)	4.23" (107 mm)	2	1 x 5½" ---	1.18" (30 mm)	⅜" (9.5 mm)	4	69.5 lbs (31.6 Kgs)
24" 600	24.000" (609.6 mm)	400 psi (28.0 bar)	180860 lbs (816.80 kN)	0-0.375" (0-9.5 mm)	0.80°	0.17"/ft (14 m/mm)	27.17" (690 mm)	30.00" (762 mm)	4.84" (123 mm)	4	⅞ x 6½" ---	1.18" (30 mm)	½" (12.7 mm)	4	101.9 lbs (46.3 Kgs)
26" 650	26.000" (660.4 mm)	300 psi (20.0 bar)	159190 lbs (684.72 kN)	0-0.500" (0-12.7 mm)	1.06°	0.22"/ft (18 m/mm)	29.58" (751 mm)	32.78" (832 mm)	6.69" (170 mm)	4	1 x 10" ---	1.97" (50 mm)	½" (12.7 mm)	4	173.5 lbs (78.7 Kgs)

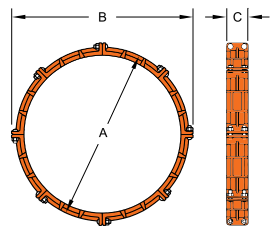
1) Dimensions are subject to change without notice. Other sizes are available on request.
* Working Pressure is based on rings both sides fully welded standard wall carbon steel pipe.
# Working Pressure and End Load are the total from all internal and external loads based on the applicable pipe wall thickness.
† Allowable Axial Displacement and Angular Movement (Deflection) figures shown are the maximum nominal range of movement at each R-88 coupling joint when rings are welded in the standard position. For design and installation purposes these figures should be reduced by 25%.
‡ The number of ring clamps listed is the minimum required to correctly position the weld ring around the circumference of the pipe end.

R-88 ring joint coupling
28" - 96"

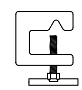
R-88
28" - 96"



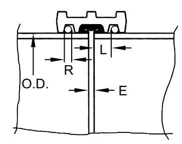
28" - 38"



40" - 96"



Ring Clamp



The Shurjoint Model R-88 Ring Joint Coupling is available in sizes 28" to 96" (700 mm to 2400 mm). The larger diameter couplings are comprised of 4 to 8 housing segments depending on the size and feature two bolts at each joint segment to ensure a positive connection.

dimensions

R88 (28" - 42")

		rings both sides fully welded**			angular movement / deflection		dimensions			bolts					
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)*	axial displacement[1] E	per coupling (°)	per pipe	A	B	C	no.	size	sealing surface L	ring size R	no. of clamps	weight
28" 700	28.0" (711.2 mm)	300 psi (20.0 bar)	184630 lbf (794.11 kN)	0-0.500" (0-12.7 mm)	0.90°	0.19"/ft (16 mm/m)	31.75" (806 mm)	35.47" (901 mm)	6.69" (170 mm)	12	⅞ x 4"	2.00" (50 mm)	½" (12.7 mm)	4	222.2 lbs (101 kg)
30" 750	30.0" (762.0 mm)	300 psi (20.0 bar)	211950 lbf (911.61 kN)	0-0.500" (0-12.7 mm)	0.86°	0.18"/ft (15 mm/m)	33.75" (857 mm)	37.6" (955 mm)	6.69" (170 mm)	12	1 x 3½"	2.00" (50 mm)	½" (12.7 mm)	4	218.9 lbs (99.5 kg)
32" 800	32.0" (812.8 mm)	300 psi (20.0 bar)	241150 lbf (1037.21 kN)	0-0.500" (0-12.7 mm)	0.84°	0.18"/ft (15 mm/m)	35.75" (908 mm)	39.45" (1002 mm)	6.69" (170 mm)	12	1 x 3½"	2.00" (50 mm)	½" (12.7 mm)	4	225.4 lbs (102.2 kg)
34"** 850	34.0" (863.4 mm)	300 psi (20.0 bar)	272230 lbf (1170.37 kN)	0-0.500" (0-12.7 mm)	0.84°	0.18"/ft (15 mm/m)	37.75" (959 mm)	41.5" (1054 mm)	7.87" (200 mm)	12	1 x 3½"	2.00" (50 mm)	½" (12.7 mm)	4	253 lbs (115 kg)
36" 900	36.0" (914.4 mm)	300 psi (20.0 bar)	305200 lbf (1312.72 kN)	0-0.500" (0-12.7 mm)	0.76°	0.16"/ft (13 mm/m)	39.75" (1010 mm)	43.5" (1103 mm)	7.87" (200 mm)	12	1 x 3½"	2.00" (50 mm)	½" (12.7 mm)	4	246 lbs (111.6 kg)
38"** 950	38.0" (965.2 mm)	232 psi (16.0 bar)	262980 lbf (1170.10 kN)	0-0.500" (0-12.7 mm)	0.76°	0.16"/ft (13 mm/m)	41.75" (1060 mm)	45.5" (1156 mm)	7.87" (200 mm)	12	1 x 3½"	2.00" (50 mm)	½" (12.7 mm)	4	275 lbs (125 kg)
40" 1000	40.0" (1016.0 mm)	232 psi (16.0 bar)	291390 lbf (1296.51 kN)	0-0.625" (0-15.9 mm)	0.80°	0.17"/ft (14 mm/m)	44.69" (1135 mm)	48.39" (1229 mm)	7.87" (200 mm)	16	1 x 3½"	2.37" (60 mm)	⅝" (15.9 mm)	6	310.2 lbs (141 kg)
42" 1050	42.0" (1066.8 mm)	232" (16.0 bar)	321250 lbf (1429.41 kN)	0-0.625" (0-15.9 mm)	0.86°	0.18"/ft (15 mm/m)	46.7" (1186 mm)	50.71 (1288 mm)	7.87" (200 mm)	16	1 x 3½	2.37" (60 mm)	⅝" (15.9 mm)	6	326.9 lbs (148.6 kg)

dimensions

R88 (44" - 96")

		rings both sides fully welded**			angular movement / deflection		dimensions			bolts					
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)*	axial displacement[1] E	per coup.	per pipe	A	B	C	no.	size	sealing surface L	ring size R	no. of clamps	weight
44"** 1100	44.0" (1117.6 mm)	232 psi (16.0 bar)	352580 lbs (1568.78 kN)	0-0.625" (0-15.9 mm)	0.80°	0.17"/ft (14 mm/m)	48.66" (1236 mm)	52.64" (1337 mm)	7.87" (200 mm)	16	1¼ x 5"	2.37" (60 mm)	⅝" (15.9 mm)	6	343.2 lbs (156 Kg)
48" 1200	48.0" (1219.2 mm)	232 psi (16.0 bar)	419600 lbs (1866.98 kN)	0-0.625" (0-15.9 mm)	0.70°	0.15"/ft (12 mm/m)	52.68" (1338 mm)	55.91" (1420 mm)	7.87" (200 mm)	16	1 x 3½"	2.37" (60 mm)	⅝" (15.9 mm)	6	466.7 lbs (211.8 Kg)
52"** 1300	52.0" (1320.8 mm)	175 psi (12.0 bar)	371460 lbs (1643.33 kN)	0-0.625" (0-15.9 mm)	---	--- ---	61.25" (1555 mm)	60.67" (1541 mm)	7.87" (200 mm)	16	1¼ x 5"	2.37" (60 mm)	⅝" (15.9 mm)	6	453.2 lbs (206 Kg)
54"** 1350	54.0" (1371.6 mm)	175 psi (12.0 bar)	400580 lbs (1772.17 kN)	0-0.625" (0-15.9 mm)	---	--- ---	63.25" (1607 mm)	62.52" (1588 mm)	7.87" (200 mm)	16	1¼ x 5"	2.37" (60 mm)	⅝" (15.9 mm)	6	472.1 lbs (214.6 Kg)
56"** 1400	56.0" (1422.4 mm)	175 psi (12.0 bar)	430800 lbs (1905.87 kN)	0-0.625" (0-15.9 mm)	---	--- ---	65.38" (1660 mm)	64.69" (1643 mm)	7.87" (200 mm)	16	1¼ x 5"	2.37" (60 mm)	⅝" (15.9 mm)	6	488.2 lbs (222 Kg)
60"** 1500	60.0" (1524.0 mm)	175 psi (12.0 bar)	494550 lbs (2187.87 kN)	0-0.625" (0-15.9 mm)	---	--- ---	69.38" (1762 mm)	68.82" (1748 mm)	7.87" (200 mm)	16	1¼ x 5"	2.37" (60 mm)	⅝" (15.9 mm)	6	537.2 lbs (244.2 Kg)
66"** 1650	66.0" (1676.4 mm)	175 psi (12.0 bar)	598709 lbs (2663.19 kN)	0-0.750" (0-19.1 mm)	---	--- ---	76.00" (1932 mm)	75.75" (1924 mm)	8" (216 mm)	16	1¼ x 5"	2.37" (60 mm)	¾" (19.1 mm)	8	612.5 lbs (278.4 Kg)
68" 1700	68.0" (1727.2 mm)	175 psi (12.0 bar)	635544 lbs (2827.04 kN)	0-0.750" (0-19.1 mm)	---	--- ---	78.50" (1994 mm)	78.03" (1982 mm)	8" (216 mm)	16	1¼ x 5"	2.37" (60 mm)	¾" (19.1 mm)	8	785.4 lbs (357 Kg)
72" 1800	72.0" (1828.8 mm)	175 psi (12.0 bar)	712513 lbs (3169.41 kN)	0-0.750" (0-19.1 mm)	---	--- ---	82.50" (2095 mm)	82.28" (2090 mm)	8" (216 mm)	16	1½ x 6⅞"	2.37" (60 mm)	¾" (19.1 mm)	8	737.7 lbs (335.3 Kg)
84"** 2100	84.0" (2133.6 mm)	100 psi (7.0 bar)	553890 lbs (2501.46 kN)	0-0.750" (0-19.1 mm)	---	--- ---	94.75" (2406 mm)	93.81" (2383 mm)	8" (216 mm)	16	1¼ x 5"	2.37" (60 mm)	¾" (19.1 mm)	8	780.3 lbs (354.7 Kg)
96"** 2400	96.0" (2438.4 mm)	100 psi (7.0 bar)	723450 lbs (3267.21 kN)	0-0.750" (0-19.1 mm)	---	--- ---	106.75" (2711 mm)	106.54" (2706 mm)	8" (216 mm)	16	1¼ x 5"	2.37" (60 mm)	¾" (19.1 mm)	8	823.2 lbs (374.2 Kg)

Dimensions are subject to change without notice. Other sizes are available on request.
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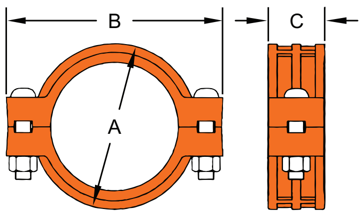
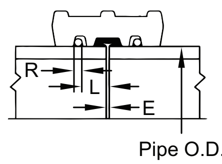
\# Working Pressure and End Load are the total from all internal and external loads based on the applicable pipe wall thickness.
† Allowable Axial Displacement and Angular Movement (Deflection) figures shown are the maximum nominal range of movement at each R-88 coupling joint when rings are welded in the standard position. For design and installation purposes these figures should be reduced by 25%.
‡ The number of ring clamps listed is the minimum required to correctly position the weld ring around the circumference of the pipe end.
** Non-standard/stock items may require longer lead time.

RH-1000 ring joint coupling
1000 psi





The Shurjoint Model RH-1000 coupling is a high pressure ring joint coupling for use with Sch. 40, Sch. 80 and heavier wall carbon steel pipelines. The coupling is comprised of two ductile iron heavy-wall housings, rubber gasket (EPDM or Nitrile) and two heat-treated track bolts and nuts and provides a fully restrained joint with maximum working pressure up to 1,000 psi (70 Bar) depending on the pipe used.

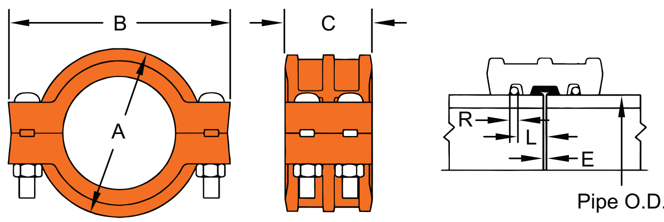
Two steel weld rings will be factory supplied with a coupling. Steel rings shall be always fully welded at both sides.

dimensions

				dimensions			bolts / nuts[1]			pipe-end preparation			
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)	A	B	C	no.	size	deflection	R	l	E max	weight
8" 200	8.625" (219.1 mm)	1000 psi (69 bar)	58390 lbs (263.79 kN)	11.10" (282 mm)	14.65" (372 mm)	3.86" (98 mm)	2	1" x 5½"	0° – 18′	0.472-0.500" (12.0-12.7 mm)	1" 25	0.13" (3.2 mm)	39.8" (18.1 mm)
10" 250	10.750" (273.0 mm)	1000 psi (69 bar)	90710 lbs (409.54 kN)	13.32" (340 mm)	16.93" (430 mm)	4.25" (108 mm)	2	1" x 6½"	0° – 38′	0.472-0.500" (12.0-12.7 mm)	1" 25	0.13" (3.2 mm)	57.2" (26.0 mm)
12" 300	12.750" (323.9 mm)	1000 psi (69 bar)	127610 lbs (576.49 kN)	16.33" (415 mm)	20.07" (510 mm)	4.17" (106 mm)	2	1" x 6½"	0° – 32′	0.472-0.500" (12.0-12.7 mm)	1" 25	0.13" (3.2 mm)	72.6" (33.0 mm)

*Working pressure is based on standard wall carbon steel pipe.
1 Bolts & nuts are UNC threaded.

RX-3000 ring joint coupling
3000 psi



The Shurjoint Model RX-3000 coupling is a high pressure ring joint coupling for use with Sch. 80, 120 or heavier wall carbon steel pipelines. The coupling is comprised of two ductile iron heavy-wall housings, rubber gasket (EPDM or Nitrile) and two or four heat-treated track bolts and nuts which provide a fully restrained joint with maximum working pressure up to 3,000 psi (210 Bar) depending on the pipe used.

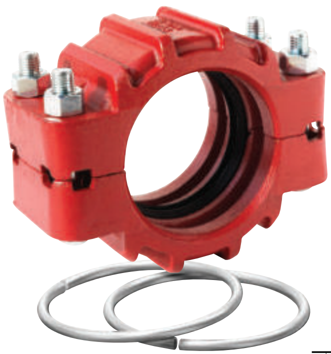
Two steel weld rings will be factory supplied with a coupling. Steel rings shall be always fully welded at both sides.

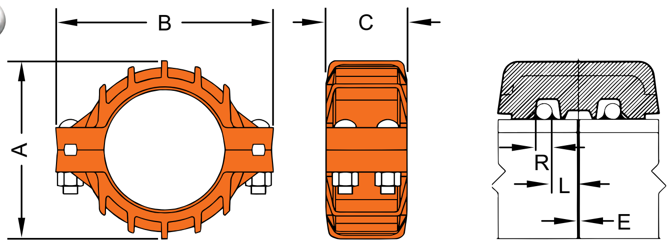
dimensions

				dimensions			bolts / nuts[1]		pipe-end preparation			
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)	A	B	C	no.	size	R	L	E	weight
8" 200	8.625" (219.1 mm)	3000 psi (207 bar)	175180 lbs (791.36 kN)	11.81" (300 mm)	15.51" (394 mm)	5.83" (148 mm)	2	1⅛ x 5⅛"	0.472-0.500" (12.0-12.7 mm)	1.22" (31 mm)	⅛" (3 mm)	78.92 lbs (35.87 kg)
10" 250	10.748" (273.0 mm)	3000 psi (207 bar)	272040 lbs (1228.61 kN)	14.96" (380 mm)	18.93" (481 mm)	5.98" (152 mm)	4	1¼ x 6½"	0.625-0.629" (15.9-16.0 mm)	1.22" (31 mm)	⅛" (3 mm)	116.36 lbs (52.78 kg)
12" 300	12.752" (323.9 mm)	3000 psi (207 bar)	382950 lbs (1729.46 kN)	18.5" (470 mm)	22.48" (572 mm)	6.81" (173 mm)	4	1½ x 6¼"	0.625-0.629" (15.9-16.0 mm)	1.22" (31 mm)	⅛" (3 mm)	212.27 lbs (96.24 kg)

*Working pressure is based on API 5L X65 line pipe.
1 Bolts & nuts are UNC threaded.

RX-3770 ring joint coupling
3770 psi





The Shurjoint Model RX-3770 Ring Joint Coupling is designed to provide a fully restrained joint for use with extra-strong carbon steel pipe including API 5L Grade X65 line pipe. The coupling is comprised of two ductile iron heavy-wall housing segments, rubber gasket (EPDM) and four heat-treated track bolts and nuts. Two steel weld rings are factory supplied with a coupling. Steel rings shall always be fully welded on both sides.

dimensions

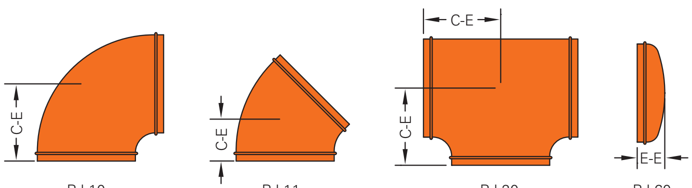
				dimensions			bolts / nuts[1]		pipe-end preparation			
nom. size	pipe o.d.	max working pressure (cwp)*	max end load (cwp)	A	B	C	no.	size	R	L	E	weight
6” 150	6.625” (168.3 mm)	3770 psi (260 bar)	129890 lbs (578.11 kN)	10.24” (260 mm)	12.64” (321 mm)	5.87” (149 mm)	4	⅞ x 6½”	0.472” (12 mm)	1.22” (31 mm)	0.20” (5 mm)	61.2 lbs (27.7 Kgs)
8” 200	8.625” (219.1 mm)	3770 psi (260 bar)	220150 lbs (979.78 kN)	12.95” (329 mm)	16.30” (414 mm)	6.89” (175 mm)	4	1¼ x 6½	0.625” (16 mm)	1.50” (38 mm)	0.20” (5 mm)	110.0 lbs (49.9 Kgs)
10” 250	10.750” (273.0 mm)	3770 psi (260 bar)	342000 lbs (1521.14 kN)	15.90” (404 mm)	19.84” (504 mm)	7.72” (196 mm)	4	1½ x 6⅞”	0.750” (19 mm)	1.50” (38 mm)	0.20” (5 mm)	174.5 lbs (79.2 Kgs)
12” 300	12.750” (323.9 mm)	3770 psi (260 bar)	481090 lbs (2141.24 kN)	19.00” (482 mm)	23.10” (587 mm)	8.63” (219 mm)	4	1½ x 6⅞”	0.875” (22 mm)	1.50” (38 mm)	0.24” (6 mm)	247.1 lbs (112.3 Kgs)

*Working pressure is based on API 5L X65 line pipe.
1 Bolts & nuts are UNC threaded.

RJ-10 90° elbow
RJ-11 45° elbow
RJ-20 tee
RJ-60 cap

full range offerings

- 8" - 16" (200 mm – 400 mm) Models RJ-10, RJ-11 & RJ-60 and Model RJ-20 are available in cast ductile iron to ASTM A536 Gr. 65-45-12.
- Larger sizes are made of carbon steel standard weight pipe, ASTM A53 Gr. B or equivalent, or fabricated from wrought carbon steel of the equivalent properties.
- Other configurations are also available upon request. Contact Shurjoint for details.



RJ-10 RJ-11 RJ-20 RJ-60

Shurjoint offers a full range of ring joint fittings for use with Model R-88 ring joint couplings.

dimensions

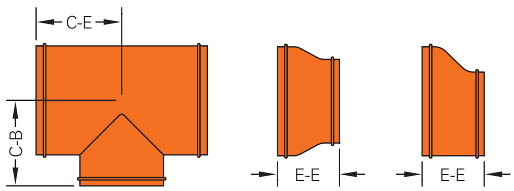
		RJ-10 90° elbow		RJ-11 45° elbow		RJ-20 tee		RJ-60 cap	
nom. size	pipe O.D.	C - E	weight	C - E	weight	C -E	weight	E - E	weight
8" 200	8.625" (219.1 mm)	7.75" (197 mm)	28.6 lbs (13.0 Kgs)	4.25" (108 mm)	20.9 lbs (9.5 Kgs)	7.75" (197 mm)	46.2 lbs (21.0 Kgs)	3.00" (76 mm)	12.1 lbs (5.5 Kgs)
10" 250	10.750" (273.0 mm)	9.00" (229 mm)	55.0 lbs (25.0 Kgs)	4.75" (121 mm)	39.6 lbs (18.0 Kgs)	9.00" (229 mm)	72.6 lbs (33.0 Kgs)	3.00" (76 mm)	13.2 lbs (6.0 Kgs)
12" 300	12.750" (323.9 mm)	10.00" (254 mm)	77.0 lbs (35.0 Kgs)	5.25" (133 mm)	50.6 lbs (23.0 Kgs)	10.00" (254 mm)	103.4 lbs (47.0 Kgs)	3.00" (76 mm)	17.6 lbs (8.0 Kgs)
200 JIS	8.516" (216.3 mm)	7.75" (197 mm)	28.6 lbs (13.0 Kgs)	4.25" (108 mm)	20.9 lbs (9.5 Kgs)	7.75" (197 mm)	46.2 lbs (21.0 Kgs)	3.00" (76 mm)	12.1 lbs (5.5 Kgs)
250 JIS	10.528" (267.4 mm)	9.00" (229 mm)	55.0 lbs (25.0 Kgs)	4.75" (121 mm)	39.6 lbs (18.0 Kgs)	9.00" (229 mm)	72.6 lbs (33.0 Kgs)	3.00" (76 mm)	13.2 lbs (6.0 Kgs)
300 JIS	12.539" (318.5 mm)	10.00" (254 mm)	77.0 lbs (35.0 Kgs)	5.25" (133 mm)	50.6 lbs (23.0 Kgs)	10.00" (254 mm)	103.4 lbs (47.0 Kgs)	3.00" (76 mm)	17.6 lbs (8.0 Kgs)
14" 350	14.000" (355.6 mm)	21.00" (533 mm)	81.4 lbs (37.0 Kgs)	6.00" (152 mm)	52.8 lbs (24.0 Kgs)	11.00" (280 mm)	118.8 lbs (54.0 Kgs)	4.00" (102 mm)	26.4 lbs (12.0 Kgs)
16" 400	16.000" (406.4 mm)	24.00" (610 mm)	99.0 lbs (45.0 Kgs)	7.25" (184 mm)	101.2 lbs (46.0 Kgs)	12.00" (305 mm)	154.0 lbs (70.0 Kgs)	4.00" (102 mm)	33.0 lbs (15.0 Kgs)
18" 450	18.000" (457.2 mm)	27.00" (686 mm)	209.0 lbs (95.0 Kgs)	11.25" (286 mm)	105.6 lbs (48.0 Kgs)	13.50" (343 mm)	268.0 lbs (122.0 Kgs)	5.00" (127 mm)	46.2 lbs (21.0 Kgs)
20" 500	20.000" (508.0 mm)	30.00" (762 mm)	203.6 lbs (138.0 Kgs)	12.50" (318 mm)	110.0 lbs (50.0 Kgs)	17.25" (438 mm)	337.0 lbs (153.0 Kgs)	6.00" (152 mm)	57.2 lbs (26.0 Kgs)
24" 600	24.000" (609.6 mm)	36.00" (914 mm)	485.0 lbs (220.0 Kgs)	15.00" (381 mm)	176.0 lbs (80.0 Kgs)	17.00" (432 mm)	466.0 lbs (212.0 Kgs)	6.00" (152 mm)	77.0 lbs (35.0 Kgs)
26" 650	26.000" (660.4 mm)	39.00" (991 mm)	521.0 lbs (237.0 Kgs)	16.00" (406 mm)	262.0 lbs (119.0 Kgs)	22.50" (572 mm)	766.0 lbs (348.0 Kgs)	10.50" (267 mm)	110.0 lbs (50.0 Kgs)
28" 700	28.000" (711.2 mm)	42.00" (1067 mm)	605.0 lbs (275.0 Kgs)	17.25" (438 mm)	304.0 lbs (138.0 Kgs)	23.50" (597 mm)	862.0 lbs (392.0 Kgs)	10.50" (267 mm)	123.0 lbs (56.0 Kgs)
30" 750	30.000" (76.20 mm)	45.00" (1143 mm)	695.0 lbs (316.0 Kgs)	18.50" (480 mm)	304.0 lbs (138.0 Kgs)	25.00" (635 mm)	992.0 lbs (451.0 Kgs)	10.50" (267 mm)	136.0 lbs (62.0 Kgs)
32" 800	32.000" (812.8 mm)	48.00" (1219 mm)	792.0 lbs (360.0 Kgs)	19.75" (502 mm)	396.0 lbs (180.0 Kgs)	26.50" (673 mm)	1135.0 lbs (516.0 Kgs)	10.50" (267 mm)	248.6 lbs (113.0 Kgs)

graph continued on next page

nom. size	pipe O.D.	RJ-10 90° elbow		RJ-11 45° elbow		RJ-20 tee		RJ-60 cap	
		C - E	weight	C - E	weight	C -E	weight	E - E	weight
34" 850	34.000" (863.4 mm)	51.00" (1295 mm)	895.0 lbs (407.0 Kgs)	21.00" (533 mm)	449.0 lbs (204.0 Kgs)	28.00" (711 mm)	1285.0 lbs (584.0 Kgs)	10.50" (267 mm)	165.0 lbs (75.0 Kgs)
36" 900	36.000" (914.4 mm)	54.00" (1372 mm)	1005.0 lbs (457.0 Kgs)	22.25" (565 mm)	504.0 lbs (229.0 Kgs)	30.00" (762 mm)	1445.0 lbs (657.0 Kgs)	10.50" (267 mm)	334.4 lbs (152.0 Kgs)
40" 1000	40.000" (1016.0 mm)	60.00" (1524 mm)	1241.0 lbs (564.0 Kgs)	24.88" (632 mm)	620.0 lbs (282.0 Kgs)	33.00" (838 mm)	1790.0 lbs (814.0 Kgs)	12.00" (305 mm)	224.0 lbs (102.0 Kgs)
42" 1050	42.000" (1066.8 mm)	63.00" (1600 mm)	1368.0 lbs (622.0 Kgs)	26.00" (660 mm)	684.0 lbs (311.0 Kgs)	35.00" (889 mm)	1841.0 lbs (837.0 Kgs)	12.00" (305 mm)	242.0 lbs (110.0 Kgs)
44" 1100	44.000" (1117.6 mm)	66.00" (1676 mm)	1503.0 lbs (683.0 Kgs)	27.39" (696 mm)	752.0 lbs (342.0 Kgs)	36.00" (914 mm)	2075.0 lbs (943.0 Kgs)	13.50" (343 mm)	277.0 lbs (126.0 Kgs)
48" 1200	48.000" (1219.2 mm)	72.00" (1829 mm)	1790.0 lbs (814.0 Kgs)	29.88" (759 mm)	895.0 lbs (407.0 Kgs)	40.00" (1016 mm)	2488.0 lbs (1131.0 Kgs)	13.50" (343 mm)	315.0 lbs (143.0 Kgs)

- C-E of RJ-10 and RJ-11 18" and larger sizes and E-E of RJ-60 26" and larger sizes conform to ANSI B16.9. All other sizes are to manufacturer's standard.

RJ-21 reducing tee
RJ-50 concentric reducer
RJ-51 eccentric reducer



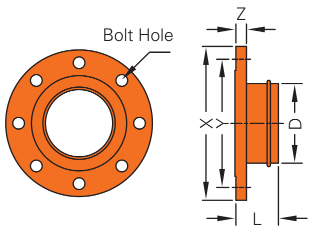
RJ-21 RJ-50 RJ-51

dimensions

nom. size	pipe O.D.	RJ-21 reducing tee			RJ-50 conc. reducer		RJ-51 ecc. reducer	
		C - E	C-B	weight	E - E	weight	E - E	weight
14 x 12” 350 x 300	14.000 x 12.750” (355.6 x 323.9 mm)	11.00” (279 mm)	10.62” (270 mm)	145.0 lbs (66.0 Kgs)	8.00” (203 mm*)	51.0 lbs (23.0 Kgs)	13.00” (330 mm)	51.0 lbs (23.0 Kgs)
16 x12” 400 x 300	16.000 x 12.750” (406.4 x 323.9 mm)	12.00” (305 mm)	11.62” (295 mm)	172.0 lbs (78.0 Kgs)	9.00” * (229 mm)	64.0 lbs (29.0 Kgs)	9.00” * (229 mm)	64.0 lbs (29.0 Kgs)
16 x 14” 400 x 350	16.000 x 14.000” (406.4 x 355.6 mm)	12.00” (305 mm)	12.00” (305 mm)	176.0 lbs (80.0 Kgs)	9.00” * (229 mm)	64.0 lbs (29.0 Kgs)	9.00” * (229 mm)	64.0 lbs (29.0 Kgs)
18 x 12” 450 x 300	18.000 x 12.750” (457.2 x 323.9 mm)	13.50” (343 mm)	12.62” (321 mm)	246.0 lbs (112.0 Kgs)	9.50” * (241 mm)	72.6 lbs (33.0 Kgs)	15.00” (381 mm)	78.0 lbs (35.0 Kgs)
18 x 14” 450 x 350	18.000 x 14.000” (457.2 x 355.6 mm)	13.50” (343 mm)	13.00” (330 mm)	253.0 lbs (115.0 Kgs)	15.00” (381 mm)	79.0 lbs (36.0 Kgs)	15.00” (381 mm)	79.0 lbs (36.0 Kgs)
18 x 16” 450 x 400	18.000 x 16.000” (457.2 x 406.4 mm)	13.50” (343 mm)	13.00” (330 mm)	264.0 lbs (120.0 Kgs)	15.00” (381 mm)	79.0 lbs (36.0 Kgs)	15.00” (381 mm)	79.0 lbs (36.0 Kgs)
20 x 12” 500 x 300	20.000 x 12.750” (508.0 x 323.9 mm)	15.00” (381 mm)	13.62” (346 mm)	297.0 lbs (135.0 Kgs)	10.00” * (254 mm)	95.0 lbs (43.0 Kgs)	20.00” (508 mm)	95.0 lbs (43.0 Kgs)
20 x 14” 500 x 350	20.000 x 14.000” (508.0 x 355.6 mm)	15.00” (381 mm)	14.00” (356 mm)	304.0 lbs (138.0 Kgs)	20.00” (508 mm)	99.0 lbs (45.0 Kgs)	20.00” (508 mm)	99.0 lbs (45.0 Kgs)
20 x 16” 500 x 400	20.000 x 16.000” (508.0 x 406.4 mm)	15.00” (381 mm)	14.00” (356 mm)	317.0 lbs (144.0 Kgs)	20.00” (508 mm)	101.0 lbs (46.0 Kgs)	20.00” (508 mm)	101.0 lbs (46.0 Kgs)
20 x 18” 500 x 450	20.000 x 18.000” (508.0 x 457.2 mm)	15.00” (381 mm)	14.50” (368 mm)	328.0 lbs (149.0 Kgs)	20.00” (508 mm)	128.0 lbs (58.0 Kgs)	20.00” (508 mm)	128.0 lbs (58.0 Kgs)
24 x 12” 600 x 300	24.000 x 12.750” (609.6 x 323.9 mm)	17.00” (432 mm)	15.62” (397 mm)	396.0 lbs (180.0 Kgs)	12.00” * (305 mm)	154.0 lbs (70.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 kgs)
24 x 14” 600 x 350	24.000 x 14.000” (609.6 x 355.6 mm)	17.00” (432 mm)	16.00” (406 mm)	407.0 lbs (185.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 kgs)
24 x 16” 600 x 400	24.000 x 16.000” (609.6 x 406.4 mm)	17.00” (432 mm)	16.00” (406 mm)	418.0 lbs (190.0 Kgs)	12.00” * (305 mm)	154.0 lbs (70.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 kgs)
24 x 18” 600 x 450	24.000 x 18.000” (609.6 x 457.2 mm)	17.00” (432 mm)	16.50” (419 mm)	433.0 lbs (197.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 Kgs)	20.00” (508 mm)	154.0 lbs (70.0 kgs)
24 x 20” 600 x 500	24.000 x 20.000” (609.6 x 508.0 mm)	17.00” (432 mm)	17.00” (432 mm)	444.0 lbs (202.0 Kgs)	20.00” (508 mm)	156.0 lbs (71.0 Kgs)	20.00” (508 mm)	156.0 lbs (71.0 Kgs)

C-E: Mfr's standard. E-E marked (*): Mfr's standard (made of ductile iron). All other E-E: ANSI B16.9.

RJ-70 flange adapter
ANSI class 125/150



RJ-70

dimensions

nom. size	pipe O.D. D	X	Y	Z	bolt size	bolt hole diameter	bolt hole no.	L	weight
8" 200	8.625" (219.1 mm)	13.500" (343.0 mm)	11.750" (298.0 mm)	1.125" (29.0 mm)	¾"	⅞"	8	6" (152 mm)	44.9 lbs (20.4 Kgs)
10" 250	10.750" (273.0 mm)	16.000" (406.4 mm)	14.250" (362.0 mm)	1.180" (30.0 mm)	⅞"	1"	12	8" (203 mm)	67.1 lbs (30.5 Kgs)
12" 300	12.750" (323.9 mm)	19.000" (483.0 mm)	17.000" (432.0 mm)	1.250" (32.0 mm)	⅞"	1"	12	8" (203 mm)	98.1 lbs (44.6 Kgs)
14" 350	14.000" (355.6 mm)	21.000" (533.0 mm)	18.750" (476.25 mm)	1.377" (35.0 mm)	1"	1⅛"	12	8" (203 mm)	118.8 lbs (54.0 Kgs)
16" 400	16.000" (406.4 mm)	23.500" (597.0 mm)	21.250" (539.75 mm)	1.456" (37.0 mm)	1"	1⅛"	16	8" (203 mm)	147.0 lbs (66.8 Kgs)
18" 450	18.000" (457.2 mm)	25.000" (635.0 mm)	22.751" (577.9 mm)	1.059" (26.9 mm)	1⅛"	1¼"	16	8" (203 mm)	143.0 lbs (65.0 Kgs)
20" 500	20.000" (508.0 mm)	27.519" (699.0 mm)	25.000" (635.0 mm)	1.692" (43.0 mm)	1⅛"	1¼"	20	8" (203 mm)	169.4 lbs (77.0 Kgs)
24" 600	24.000" (609.6 mm)	32.031" (813.6 mm)	29.500" (749.3 mm)	1.889" (48.0 mm)	1¼"	1⅜"	20	8" (203 mm)	286.9 lbs (130.4 Kgs)

L: Mfr's standard.

pressure performance data

The following tables show maximum cold working pressures (CWP) of Shurjoint R-88 couplings based on rings both sides fully welded and corresponding working pressure for applicable steel pipe.

		max. working pressure / max. end load rings both sides fully weilded					
nom. size	**pipe O.D.**	**XS (.500”)**		**STD (.375”)**		**LW (.312”)**	
8” 200	8.625” (219.1 mm)	600 psi (42 bar)	35040 lbs (150.74 kN)	400 psi (28 bar)	23359 lbs (105.51 kN)	400 psi (28 bar)	23359 lbs (105.51 kN)
10” 250	10.750” (273.0 mm)	600 psi (42 bar)	54430 lbs (234.02 kN)	400 psi (28 bar)	36287 lbs (163.81 kN)	400 psi (28 bar)	36287 lbs (163.81 kN)
12” 300	12.750” (323.9 mm)	600 psi (42 bar)	76567 lbs (329.42 kN)	400 psi (28 bar)	51045 lbs (230.59 kN)	400 psi (28 bar)	51045 lbs (230.59 kN)
200 JIS	8.516” (216.3 mm)	600 psi (42 bar)	34215 lbs (150.58 kN)	400 psi (28 bar)	22772 lbs (102.83 kN)	400 psi (28 bar)	22772 lbs (102.83 kN)
250 JIS	10.528” (267.4 mm)	600 psi (42 bar)	52205 lbs (224.52 kN)	400 psi (28 bar)	34803 lbs (157.16 kN)	400 psi (28 bar)	34803 lbs (157.16 kN)
300 JIS	12.539” (318.5 mm)	600 psi (42 bar)	74054 lbs (318.53 kN)	400 psi (28 bar)	49369 lbs (222.97 kN)	400 psi (28 bar)	49369 lbs (222.97 kN)
14” 350 (R-88N)	14.000” (355.6 mm)	600 psi (42 bar)	92316 lbs (397.06 kN)	400 psi (28 bar)	61544 lbs (277.94 kN)	350 psi (24 bar)	53851 lbs (238.23 kN)
16” 400 (R-88N)	16.000” (406.4 mm)	500 psi (35 bar)	100480 lbs (453.78 kN)	400 psi (28 bar)	80384 lbs (363.02 kN)	350 psi (24 bar)	70336 lbs (311.16 kN)
18” 450 (R-88N)	18.000” (457.2 mm)	500 psi (35 bar)	12170 lbs (574.31 kN)	400 psi (28 bar)	101736 lbs (459.45 kN)	350 psi (24 bar)	89019 lbs (393.82 kN)
20” 500(R-88N)	20.000” (508.0 mm)	500 psi (35 bar)	157000 lbs (709.03 kN)	400 psi (28 bar)	125600 lbs (567.22 kN)	300 psi (20 bar)	94200 lbs (405.16 kN)
24” 600 (R-88N)	24.000” (609.6 mm)	500 psi (35 bar)	226080 lbs (1021.00 kN)	400 psi (28 bar)	180864 lbs (816.80 kN)	250 psi (17 bar)	113040 lbs (495.92 kN)
26” 650 (R-88N)	26.000” (660.4 mm)	400 psi (28 bar)	212264 lbs (958.61 kN)	300 psi (20 bar)	159198 lbs (584.72 kN)	250 psi (17 bar)	132665 lbs (582.01 kN)
28” 700	28.000” (711.2 mm)	400 psi (28 bar)	246176 lbs (1111.76 kN)	300 psi (20 bar)	184632 lbs (794.11 kN)	250 psi (17 bar)	153860 lbs (675.00 kN)
30” 750	30.000” (762.0 mm)	400 psi (28 bar)	282600 lbs (1276.26 kN)	300 psi (20 bar)	211950 lbs (911.61 kN)	250 psi (17 bar)	176625 lbs (774.87 kN)
32” 800	32.000” (812.8 mm)	400 psi (28 bar)	321536 lbs (1452.10 kN)	300 psi (20 bar)	241152 lbs (1037.21 kN)	250 psi (17 bar)	200960 lbs (881.63 kN)
34” 850	34.000” (863.4 mm)	350 psi (24 bar)	317611 lbs (1404.45 kN)	300 psi (20 bar)	272238 lbs (1170.37 kN)	200 psi (14 bar)	181492 lbs (819.26 kN)
36” 900	36.000” (914.4 mm)	350 psi (24 bar)	356076 lbs (1575.26 kN)	300 psi (20 bar)	305208 lbs (1312.72 kN)	200 psi (14 bar)	203472 lbs (918.90 kN)
38” 950	38.000” (965.2 mm)	300 psi (20 bar)	340062 lbs (1462.63 kN)	232 psi (16 bar)	262981 lbs (1170.10 kN)	175 psi (12 bar)	198370 lbs (877.58 kN)
40” 1000	40.000” (1016.0 mm)	300 psi (20 bar)	376800 lbs (1620.64 kN)	232 psi (16 bar)	291392 lbs (1296.51 kN)	175 psi (12 bar)	219800 lbs (972.39 kN)
42” 1050	42.000” (1066.8 mm)	300 psi (20 bar)	415422 lbs (1786.76 kN)	232 psi (16 bar)	321260 lbs (1429.41 kN)	175 psi (12 bar)	242330 lbs (1072.05 kN)
44” 1100	44.000” (1117.6 mm)	300 psi (20 bar)	455928 lbs (1960.98 kN)	232 psi (16 bar)	352584 lbs (1568.78 kN)	175 psi (12 bar)	265958 lbs (1176.59 kN)
48” 1200	48.000” (1219.2 mm)	300 psi (20 bar)	542592 lbs (2333.72 kN)	232 psi (16 bar)	419604 lbs (1866.98 kN)	--- ---	--- ---
52” 1300	52.000” (1320.8 mm)	232 psi (16 bar)	492452 lbs (2191.11 kN)	175 psi (12 bar)	371462 lbs (1643.33 kN)	--- ---	--- ---
54” 1350	54.000” (1371.6 mm)	232 psi (16 bar)	531062 lbs (2362.90 kN)	175 psi (12 bar)	400586 lbs (1772.17 kN)	--- ---	--- ---
56” 1400	56.000” (1422.4 mm)	232 psi (16 bar)	571128 lbs (2541.17 kN)	175 psi (12 bar)	430808 lbs (1905.87 kN)	--- ---	--- ---

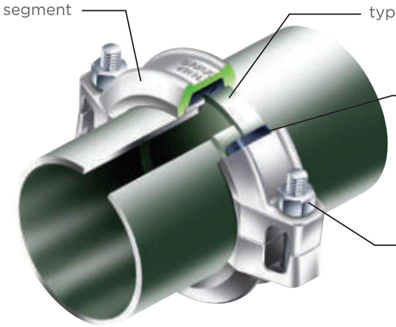
graph continued on next page

nom. size	pipe O.D.	max. working pressure / max. end load rings both sides fully weilded					
		XS (.500”)		STD (.375”)		LW (.312”)	
60” 1500	60.000” (1524.0 mm)	232 psi (16 bar)	656532 lbs (2917.16 kN)	175 psi (12 bar)	494550 lbs (2187.87 kN)	--- ---	--- ---
66” 1650	66.000” (1676.4 mm)	175 psi (12 bar)	598406 lbs (2647.32 kN)	125 psi (9 bar)	427433 lbs (1897.24 kN)	--- ---	--- ---
68” 1700	68.000” (1727.2 mm)	175 psi (12 bar)	635222 lbs (2810.19 kN)	125 psi (9 bar)	453730 lbs (2013.97 kN)	--- ---	--- ---
72” 1800	72.000” (1828.8 mm)	150 psi (10 bar)	610416 lbs (2625.44 kN)	125 psi (9 bar)	508680 lbs (2257.88 kN)	--- ---	--- ---
84” 2100	84.000” (2133.6 mm)	125 psi (9 bar)	692370 lbs (3073.22 kN)	100 psi (7 bar)	553896 lbs (2501.46 kN)	--- ---	--- ---
96” 2400	96.000” (2438.4 mm)	125 psi (9 bar)	904320 lbs (4014.01 kN)	100 psi (7 bar)	723456 lbs (3267.21 kN)	--- ---	--- ---

Shurjoint shouldered piping system

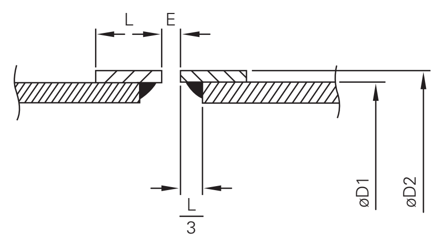
The Shurjoint shouldered piping system is a classic and versatile piping method used for a wide range of applications including irrigation, dewatering on construction sites, and other installations, etc. The shouldered system features full flow characteristics, provides speed and ease of installation, and proven reliability. The system provides for limited expansion and contraction and accommodates some linear and angular movement. Each joint is a union.

The Shurjoint shouldered piping system utilizes Type A weld-on rings, manufactured from mild steel or material compatible to pipe end used.



shoulder dimensions (type A weld-on rings)

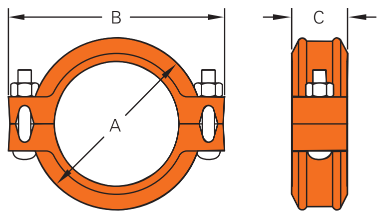
The Shurjoint shouldered piping system utilizes Type A weld-on rings, manufactured from mild steel or material compatible to pipe end used. Type A rings are suitable for services up to maximum 600 psi (40 Bar) for sizes up to 4" (100 mm) and 400 psi (28 Bar) for 5" (125 mm) through 8" (200 mm).



nom. size	pipe O.D. ϕ D1	shoulder dia. ϕ D2	shoulder length L	nom. gap E
2" 50	2.375" (60.3 mm)	2.618" (66.5 mm)	0.625" (16.0 mm)	0.125" (3.2 mm)
3" 80	3.500" (88.9 mm)	3.819" (97.0 mm)	0.625" (16.0 mm)	0.125" (3.2 mm)
4" 100	4.500" (114.3 mm)	4.803" (122.0 mm)	0.688" (17.5 mm)	0.125" (3.2 mm)
165.1 mm	6.500" (165.1 mm)	6.870" (174.5 mm)	0.625" (16.0 mm)	0.125" (3.2 mm)
6" 150	6.625" (168.3 mm)	7.007" (178.0 mm)	0.625" (16.0 mm)	0.125" (3.2 mm)
8" 200	8.625" (219.1 mm)	9.134" (232.0 mm)	0.813" (20.6 mm)	0.125" (3.2 mm)
10" 250	10.750" (273.0 mm)	11.260" (286.0 mm)	0.813" (20.6 mm)	0.125" (3.2 mm)
12" 300	12.750" (323.9 mm)	13.248" (336.5 mm)	0.813" (20.6 mm)	0.125" (3.2 mm)

Note: The exterior surface and the edge of the shouldered pipe ends must be free from any indentations, projections or other harmful surface defects such as weld splatters, any lumps of galvanizing, rust, dirt and score marks. Shouldered rings must be contacted or near tight to the pipe. The "stand off length* " must be accurately consistent in the circumference.
(*:The distance between the edge of the Shouldered ring and the pipe end (⅓ the shoulder length 'L').

S35 shouldered flexible coupling



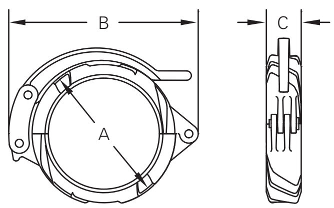
The Shurjoint Model S35 coupling is a flexible type shouldered coupling for general applications for use on Type A shouldered pipe ends. The housing segments are made of ductile iron to ASTM A536 Gr. 65-45-12 and or A395 Gr. 65-45-15 and are normally supplied in hot-dip galvanized. The standard rubber gasket is Grade T Nitrile.

dimensions

nom. size	pipe o.d.	max working pressure (cwp)*	dimensions A	dimensions B	dimensions C	allowable pipe end seperation	(°) deflection	bolt size	weight
2" 50	2.375" (60.3 mm)	600 psi (40 bar)	3.90" (99.0 mm)	5.47" (139.0 mm)	1.81" (46.0 mm)	0.13" (3.2 mm)	2° 43'	⅜" x 2⅛"	2.4 lbs (1.1 Kgs)
3" 80	3.500" (88.9 mm)	600 psi (40 bar)	5.08" (129.0 mm)	6.61" (168.0 mm)	1.81" (46.0 mm)	0.13" (3.2 mm)	1° 53'	½" x 3"	3.3 lbs (1.5 Kgs)
4" 100	4.500" (114.3 mm)	600 psi (40 bar)	6.26" (159.0 mm)	7.87" (200.0 mm)	1.97" (50.0 mm)	0.13" (3.2 mm)	1° 29'	½" x 3"	4.9 lbs (2.2 Kgs)
165.1 mm	6.500" (165.1 mm)	600 psi (40 bar)	8.39" (213.0 mm)	10.50" (267.0 mm)	1.97" (50.0 mm)	0.13" (3.2 mm)	1° 2'	⅝" x 3½"	7.7 lbs (3.5 Kgs)
6" 150	6.625" (168.3 mm)	600 psi (40 bar)	8.62" (219.0 mm)	11.00" (279.0 mm)	1.97" (50.0 mm)	0.13" (3.2 mm)	1° 1'	⅝" x 3½"	8.4 lbs (3.8 Kgs)
8" 200	8.625" (219.1 mm)	600 psi (40 bar)	10.75" (273.0 mm)	13.19" (335.0 mm)	2.36" (60.0 mm)	0.13" (3.2 mm)	0° 47'	¾" x 4¾"	13.0 lbs (5.9 Kgs)
10" 250	10.750" (273.0 mm)	300 psi (20 bar)	13.20" (335.4 mm)	15.62" (396.8 mm)	2.56" (65.0 mm)	0.13" (3.2 mm)	0° 47'	¾" x 4¾"	19.8 lbs (9.0 Kgs)
12" 300	12.750" * (323.9mm)	300 psi (20 bar)	15.19" (385.8mm)	17.72" (450.1mm)	2.56" (65.0mm)	0.13" (3.2 mm)	0° 47'	¾" x 4¾"	22.9 lbs (10.4 Kgs)

* Non-standard/stock items may require longer lead time.

SD-28A shouldered toggle coupling



The Shurjoint Model SD-28A coupling is designed to connect shouldered-end pipe with Type A rings for services where frequent assembly and disassembly is desired or required. The housing segments are hinged with a lever handle for easy installation. The use of a split pin prevents accidental opening of the couplings. The housing segments are made of ductile iron to ASTM A536 Gr. 65-45-12 and or A395 Gr. 65-45-15 and are normally supplied in hot-dip galvanized. The standard rubber gasket is Grade T Nitrile.

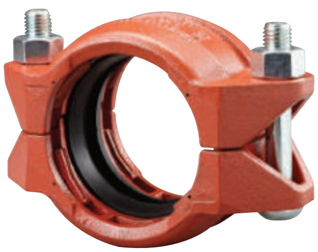
dimensions

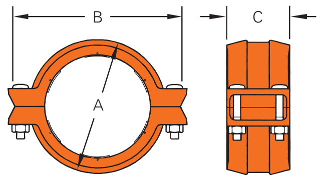
nom. size	pipe o.d.	max working pressure (cwp)*	dimensions A	dimensions B	dimensions C	allowable pipe end seperation	deflection	weight
2" * 50	2.375" (60.3 mm)	400 psi (28 bar)	39.32" (86.5 mm)	4.43" (112.5 mm)	1.93" (49 mm)	0.125" (3.2 mm)	2°-43'	2.8 lbs (1.3 Kgs)
3" * 80	3.500" (88.9 mm)	400 psi (28 bar)	4.96" (126.0 mm)	6.46" (164.0 mm)	1.93" (49 mm)	0.125" (3.2 mm)	1°-53'	4.4 lbs (2.0 Kgs)
4" 100	4.500" (114.3 mm)	400 psi (28 bar)	6.30" (160.0 mm)	8.43" (214.0 mm)	2.05" (52 mm)	0.125" (3.2 mm)	1°-29'	6.6 lbs (3.0 Kgs)
6" * 150	6.500" (165.1 mm)	400 psi (28 bar)	8.43" (214.0 mm)	11.14" (283.0 mm)	2.05" (52 mm)	0.125" (3.2 mm)	1°-2'	6.5 lbs (4.3 Kgs)
6" 150	6.625" (168.3 mm)	400 psi (28 bar)	8.54" (217.0 mm)	11.10" (282.0 mm)	2.05" (52 mm)	0.125" (3.2 mm)	1°-1'	9.9 lbs (4.5 Kgs)
8" * 200	8.625" (219.1 mm)	400 psi (28 bar)	10.95" (278.0 mm)	14.17" (360.0 mm)	2.36" (60 mm)	0.125" (3.2 mm)	0°-47'	18.3 lbs (8.3 Kgs)

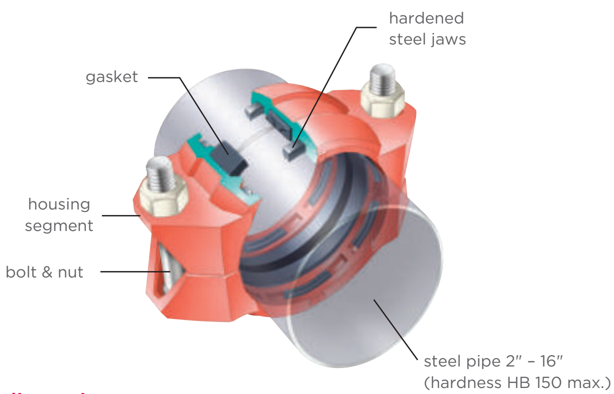
* Non-standard/stock items may require longer lead time.
** Working pressure is based on standard wall carbon steel pipe.

plain-end piping system for steel pipe

79 wildcat coupling







79 wildcat coupling

The Shurjoint Model 79 Wildcat Coupling is designed to mechanically join plain-end or beveled end carbon steel pipe. The Wildcat couplings can be used for a variety of applications including mining, process piping, manifold piping and oilfield services. The Wildcat couplings feature case-hardened jaws* within the housings and large diameter heat treated track bolts that when tightened securely grip the pipe surface. As with grooved couplings, a C-shaped rubber gasket effectively seals the pipe ends (* For sizes larger than 14" (350 mm) , jaws are made of 17-4PH stainless steel.)

The Model 79 coupling is recommended for use on carbon steel pipe with a hardness less than HB150, not recommended for stainless steel, plastic, HDPE, cast iron or other brittle pipe. Gaskets are available either in Grade E EPDM for water services of -29°F to + 230°F (-34°C to + 110°C) or Grade T Nitrile for oil services -20°F to +180°F (-29°C to +82°C).

caution: bolts and nuts must always be tightened to the required torque

dimensions

					bolt		dimensions			
nom. size	pipe o.d.	max working pressure	max end load	required bolt torque	no.	size	A	B	C	weight
1" 25	1.315" (33.4 mm)	750 psi (52 bar)	1020 lbs (4.55 kN)	150 lbs-ft (200 Nm)	2	½ x 2⅜"	2.60" (66 mm)	4.37" (111 mm)	3.05" (78 mm)	3.3 lbs (1.5 Kgs)
1½" 40	1.900" (48.3 mm)	750 psi (52 bar)	2130 lbs (9.52 kN)	150 lbs-ft (200 Nm)	2	½ x 2⅜"	3.15" (80 mm)	5.08" (129 mm)	3.05" (78 mm)	3.9 lbs (1.8 Kgs)
2" 50	2.375" (60.3 mm)	750 psi (52 bar)	3320 lbs (14.84 kN)	150 lbs-ft (200 Nm)	2	⅝ x 3½"	3.75" (95 mm)	5.94" (151 mm)	3.54" (90 mm)	7.0 lbs (3.2 Kgs)
2½" 65	2.875" (73.0 mm)	600 psi (42 bar)	3890 lbs (17.57 kN)	150 lbs-ft (200 Nm)	2	⅝ x 3½"	4.25" (108 mm)	6.46" (164 mm)	3.54" (90 mm)	7.3 lbs (3.3 Kgs)
3" 80	3.500" (88.9 mm)	600 psi (42 bar)	5770 lbs (26.06 kN)	200 lbs-ft (270 Nm)	2	¾ x 4¾"	5.00" (127 mm)	7.48" (190 mm)	3.54" (90 mm)	11.0 lbs (5.0 Kgs)
4" 100	4.500" (114.3 mm)	450 psi (31 bar)	7150 lbs (32.82 kN)	200 lbs-ft (270 Nm)	2	¾ x 4¾"	6.14" (154 mm)	8.78" (223 mm)	4.00" (102 mm)	14.3 lbs (6.5 Kgs)
5" 125	5.563" (141.3 mm)	300 psi (20 bar)	7290 lbs (32.91 kN)	250 lbs-ft (340 Nm)	2	⅞ x 6½"	7.36" (187 mm)	10.31" (262 mm)	4.38" (111 mm)	24.2 lbs (11.0 Kgs)
6" 150	6.625" (168.3 mm)	300 psi (20 bar)	10340 lbs (46.69 kN)	250 lbs-ft (340 Nm)	2	⅞ x 6½"	8.50" (216 mm)	11.50" (292 mm)	4.38" (111 mm)	28.6 lbs (13.0 Kgs)
8" 200	8.625" (219.1 mm)	250 psi (17 bar)	14600 lbs (64.06 kN)	200 lbs-ft (270 Nm)	4	¾ x 4¾"	10.88" (276 mm)	14.02" (356 mm)	5.00" (127 mm)	41.8 lbs (19.0 Kgs)
10" 250	10.750" (273.0 mm)	250 psi (17 bar)	22680 lbs (99.46 kN)	300 lbs-ft (400 Nm)	4	⅞ x 6½"	12.60" (320 mm)	16.14" (410 mm)	5.00" (127 mm)	52.8 lbs (24.0 Kgs)
12" 300	12.750" (323.9 mm)	250 psi (17 bar)	31900 lbs (140.00 kN)	350 lbs-ft (470 Nm)	4	1 x 6½"	14.60" (371 mm)	18.54" (471 mm)	5.00" (127 mm)	63.1 lbs (28.7 Kgs)
14" 350	14.000" (355.6 mm)	200 psi (14 bar)	30770 lbs (138.91 kN)	350 lbs-ft (470 Nm)	4	1 x 6½"	16.70" (424 mm)	20.00" (508 mm)	5.28" (134 mm)	93.5 lbs (42.5 Kgs)
16" 400	16.000" (406.4 mm)	150 psi (10 bar)	30140 lbs (129.65 kN)	350 lbs-ft (470 Nm)	4	1 x 6½"	18.70" (475 mm)	22.00" (559 mm)	5.28" (134 mm)	95.7 lbs (43.5 Kgs)

* Working pressure is for plain-end standard wall steel pipe with hardness less than HB150.